President's office
REGINAL ADMINISTRATION AND LOCAL GOVERNMENT
NAMANGA SECONDARY SCHOOL
FORM THREE MIDTERM EXAMINATION MARCH 2023
BASIC MATHEMATICS

TIME:3:00HRS

INSTRUCTINS:

I)This paper consists of section A and B
II)Attempt all questions in both sections
III)Show all working and answers clearly
IV)Write your name on every page of your answer sheet.

SECTION A(60 MARKS)
Answer **all questions** in this section

1 (a)Round of 783.4696 in,
(i)Three decimal places
(ii)Two significant figure
(iii)Ones

(b)Three bells are set to ring at intervals of 15 minutes, 20 miutes and 24 minutes. If they started together at 8:00 a.m, then find at what time will the bells ring together for the second time.

2 (a)If $(3^{x+3})(5^{2-y}) = (1/3)^5(1/5)$,find the values of x and y.
(b)Find a linear function f(x) such that m= -4 and f(2)=6 where m is a slope of the function.

3 (a)If R={(x,y): $y=x^2+8$) }, Find domain and range of R

(b)There are 30 men at a wedding.Twenty are businessmen, twelve are fishermen and 6 are both businessmen and fishermen.
(i) How many are neither businessmen nor fishermen?
(ii) How many are either businessmen or fishermen?

4 (a)Find the equation of the line whose gradient is 2 and its x-intercept is -6.

(b)Find gradient, x-intercept and y-intercept of the line whose equation is $\frac{3}{4}y - 6x + 7 = 0$

5 (a)Two angles are supplementary. One angle is six times the other. What are the angles?

(b)The interior angle of a regular polygon is 108^0 larger than the exterior angle. How many sides does the polygon has?

6 (a)Solve $28x^2 + 3x = 1$ by spilitting the middle term.

(b)If a positive number is decreased by 10, it is equal to 75 times its reciprocal. Find that number.

7 (a)If $x^2+8x+q=(x-k)^2$. Find the value of k and q.
(b)From the top of the tower, the angle of depression of a point on the ground 100m away from the base of the tower is 60^0, how high is the tower?

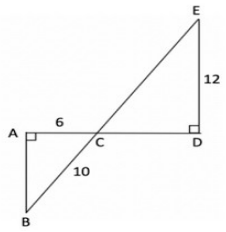
8 (a)Given $x=1.4\times10^6$ and $y=2.8\times10^4$, find z in standard form if $xz=y$

(b)Solve for x and y in.

$$2^{2x-3y} = 16$$

$$5^{x-2y} = 1$$

9 (a)In the figure below, line segments AD and BE intersect at point C. What is the length of line segment BE?



(b)Find the image of the point (4,-3) after the reflection in the line y= -x followed by another reflection in the line y-x=0.

10 (a)Find the value of x if $\log_3(x+16) - \log_3 x=2$

(b)If *p* varies directly as the square of *q* and inversely as the square root of *r*, and p=60 when q=6 and r=81, find p when q=8 and r=144.

SECTION B(40 MARKS)

Answer **all questions** in this section

11 From the table below,

Class intervals	20-25	26-31	32-37	38-43	44-49	50-55
Frequency	4	3	12	6	1	8

Evaluate the following

(i) Mean

(ii) Median

(iii) Mode

12 Given $f(x)= -4x^2-8x+3$. Find,

(i)If the graph will open downward or upward?

(ii)Axis of symmetry.

(iii)Maximum or minimum value.

(iv)Turning point.

(v)Domain and range.

13 Given that,

$$f(x)=\begin{cases} x+3 & \text{if } x<-1 \\ x^2-2 & \text{if } -1\le x<4 \\ x-4 & \text{if } x\ge 4 \end{cases}$$

(i)Sketch the graph of f(x).

(ii)state domain and range from the graph.

(iii)Evaluate f(-1), f(2) and f(5).

(iv)With a reason state whether f(x) is one to one function or not.

14 a)The function $f(x)=x^2-2x-3$, find the domain and range of inverse of this function.

b)A team of 50persons takes 12months to complete $20m^2$ stadium, how long will a team of 42persons takes to complete $35m^2$ stadium?